# Special Cases of the Traveling Salesman Problem

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ABSTRACT

The paper surveys special cases of the traveling salesman problem that are in the literature. The tables include polynomial time recognition algorithms and relationships among special cases. As a result of research, the most general special cases are identified. The paper also contains a table of subclasses of the most general special cases. Applications and open questions are indicated.

## 1. INTRODUCTION

There is current interest in special cases of the traveling salesman problem (TSP) since these algorithms find an optimal tour in polynomial time for a subset of TSPs when the number of cities is arbitrarily large. New special cases have been published recently in [5, 24, 29, 30]. Burkard surveys some of the special cases in [3] and proposes (apparently, as [6] does also) that research be directed toward approximating a given TSP with a special case so that a near optimal solution of the original TSP can be found in polynomial time. Burkard's proposed research is an extension of the application of recognizing that a given TSP is a special case.

Therefore, it seems that a survey of the most general special cases of the TSP is needed, and that recognition algorithms for special cases be pointed out or provided. The purpose of this paper is to meet these two needs. Section 2 contains information about the most general special cases, polynomial time recognition algorithms, and relationships between special cases. Several open questions are indicated. In Section 3 we identify some special



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cases that are subcases of other special cases. This classification appears to be new for most cases. The paper concludes with a 6 city 0, 1 TSP that is not in any special case.

A TSP on $n$ cities is characterized by an $n \times n$ matrix $C = [c_{ij}]$ where $c_{ij}$ may be regarded as the distance from city $i$ to city $j$. A *tour* for the salesman is a cyclic permutation on $\{1, 2, \ldots, n\}$. The length of tour $\sigma$ for $C$, denoted by $f(\sigma)$, is the sum $f(\sigma) = c_{1\sigma(1)} + c_{2\sigma(2)} + \cdots + c_{n\sigma(n)}$. A solution for distance matrix $C$ is an *optimal tour*, which is a tour $\sigma$ such that the sum $f(\sigma)$ is a minimum over all tours. Distance matrix $C$ is *symmetric* if $c_{ij} = c_{ji}$ for all $i$ and $j$. We assume that $n \geq 4$. Recall that the diagonal elements of a distance matrix have no role in the TSP and may be assigned or reassigned values for other purposes.

In order to view special cases in the context of other work on the TSP (such as [18–20]), we show broad categories of algorithms in Table 1. The first line is included to indicate the situation if the $P = NP$ question is resolved.

There is no published definition for a special case of the TSP. Table 1 helps to categorize them, along with the requirement that a special case will solve some TSPs with an arbitrarily large number of cities.

In addition to [3], there is an excellent survey of some of the special cases in [11]. Since extensive bibliographies accompany [3] and [11], many references cited there are not included in this paper.

Several results in this paper are for 0, 1 distance matrices since it is known that the 0, 1 TSP is NP-complete, and since any distance matrix can be transformed to an equivalent 0, 1 distance matrix by the method in [16], which transforms the assignment problem.

## 2. MOST GENERAL ALGORITHMS FOR SPECIAL CASES

Table 2 contains pertinent information about the most general special cases of the TSP. If a given TSP can be recognized as a special case, then a polynomial time algorithm can be used to find an optimal tour for the TSP.

TABLE 1

| Algorithm | Applicable to all TSPs? | Always yields optimal solution? | Polynomial Time? |
|---|---|---|---|
| If $NP = P$ | Yes | Yes | Yes |
| Heuristic | Yes | No | Yes |
| Exact | Yes | Yes | No |
| Special Case | No | Yes | Yes |

Therefore, data about recognition algorithms that execute in polynomial time are included in the table. Note that four special cases have open questions about a polynomial time recognition algorithm. Please see the references for the notions that are not explained in the table. This allows the survey to be brief.

TABLE 2

| Special Case | Reference | Polynomial Time Recognition Algorithm | Comment |
|---|---|---|---|
| upper triangular | [11, p. 96] | [11, p. 97] Exercise 8 results in a $O(n^5)$ algorithm | |
| Demidenko | [11, p. 102] [3, Prop. 4] | [11, pp. 102, 103] definition yields a $O(n^3)$ algorithm | Note 0 |
| generalized distribution | [5, Sec. 3.2] | [5, p. 798] definition yields a $O(n^5)$ algorithm | Note 0 |
| Fuller–Garfinkel | [11, p. 117] | definition in [11] yields a $O(n^2)$ algorithm | Note 1 |
| Gilmore–Gomory | [11, p. 125] | open question | Notes 2–4 |
| permuted distribution and tree | [11, p. 135, Thm. 22] | open question | |
| Gaikov | [11, p. 135, Cor. 8] | open question | |
| symmetric increasing | [25, Eq. (3.14)] | definition in [25] yields a $O(n^4)$ algorithm when vector $d$ is the identity | Note 5 |
| small | [26, Thm. 3] | definition in [26] yields a $O(n^2)$ algorithm | Note 6 |
| class $E$ | [1, Eq. (9)] | [1] Eq. (9) implies a $O(n^2)$ algorithm | Note 7 |
| complete | [6, p. 138 Cond. 1] | definition in [6] yields a $O(n^2)$ algorithm | |
| $N$-line planar ($N$ fixed) | [24] | a $O(n^2)$ algorithm can be based on coordinates of points and slopes of parallel lines, see [31] | |
| necklace planar | [3, p. 282] | [10, 3; Prop. 10] | |
| convex-hull-and-line | [9] | open question | |
| assignment | [4, 22, 27] | Assign large value to diagonal Use Hungarian algorithm If solution is cyclic, then it solves TSP | |
| nonoptimal arcs | [13, Thms. 1 & 2] | [13] Section 4 contains a $O(n^3)$ algorithm | |
| optimal arcs | [29, Thm. 1] | [29] Section 5 implies a $O(n^4)$ algorithm | |

TABLE 2—(*Continued*)

| Special Case | Reference | Polynomial Time Recognition Algorithm | Comment |
|---|---|---|---|
| symmetric skew orderable | [21, p. 536] | skew-$O(n^4)$ algorithm based on definition in [21, p. 536] orderable-$O(n^3)$ algorithm based on Definition 2 and construction of renumbering in [21, p. 535] | |
| symmetric band | [30, Thm. 1] | [30, proof of Theorem 1] | |
| traps | [23] | [23] Eq. (1)–(6) imply a $O(n^2)$ algorithm | |
| Brownian | [5, Sect. 4] | definition in [5] yields a $O(n^2)$ algorithm | |
| symmetric nonnegative left-upper-triangular | [5, Sect. 5] | definition in [5] yields a $O(n^2)$ algorithm | |

NOTES

0. The Demidenko and generalized distribution special cases recognize TSPs that have an optimal tour that is pyramidal. A polynomial time method to find a shortest pyramidal tour is presented in [11, p. 100] and [3, pp. 279, 280].
1. A Fuller–Garfinkel TSP, which satisfies $s_j \geq f_i$ if and only if $s_i \geq f_j$ for all $i$ and $j$ such that $i \neq j$, is a subcase of the upper triangular case. See [31].
2. If the $f_i$ and $s_i$ are increasing in a Gilmore–Gomory TSP, then it is a subcase of class $C$. See [1, 2].
3. Reference [31] contains a polynomial time recognition algorithm for 0, 1 Gilmore–Gomory TSPs.
4. Paper [7] claims to contain a recognition algorithm for Gilmore–Gomory TSPs. However, the paper has major errors that fatally flaw the approach to an algorithm. On p. 232 of [7] the partitions of $\{1, \ldots, n\}$ satisfying the conditions may not be possible for some Gilmore–Gomory TSPs, which means that Figure 1 and some other results in [7] cannot be obtained. On p. 234 in Step 2 for some Gilmore–Gomory TSPs, $\Delta$ is always equal to 0, which means that the proposed algorithm fails to recognize these Gilmore–Gomory TSPs. On p. 235 in Step 4 for some intervals, such as $A_1 < B_1 < B_2 < A_2$, the proposed algorithm does not correctly back out prior computations, which can result in faulty $f(x)$ and $g(x)$.
5. If the vector $d$ for a symmetric increasing TSP $A$ is the identity, then $-A$ is a distribution TSP.
6. A small TSP with $a_i$ decreasing and $b_j$ increasing is a subcase of class $C$. See [11, Exercise 16(d)].
7. TSP $A$ is in class $E$ if and only if $-A$ is in class $C$. Class $E$ and class 1 in [2] are identical.

## 3. RESTRICTIVE ALGORITHMS FOR SPECIAL CASES

Table 3 contains pertinent information about some of the special cases that are subcases of the special cases in Table 2. We have omitted the seven special cases in Propositions 1 and 3 of [3], which are subcases of the Demidenko special case. Also, we have omitted the five special cases in Exercise 16 of [11], which are subcases of the distribution subcase.

There are other subcases in the literature that are formed by a restriction on a special case listed in Section 2, e.g., symmetry or 0, 1 values as in [14] and [29]. These also are omitted from Table 3.

## 4. EXAMPLE

The following TSP is not in any special case. This example points out the need to understand why certain TSPs are not in any special case. The optimal tours are (125634), (126543), (136254), and their inverses. They have length 0.

$$\begin{bmatrix} — & 0 & 0 & 0 & 1 & 1 \\ 0 & — & 1 & 1 & 0 & 0 \\ 0 & 1 & — & 0 & 1 & 0 \\ 0 & 1 & 0 & — & 0 & 1 \\ 1 & 0 & 1 & 0 & — & 0 \\ 1 & 0 & 0 & 1 & 0 & — \end{bmatrix}$$

TABLE 3

| Special Case | Reference | Relationship to Other Special Cases |
|---|---|---|
| constant tour | [11, p. 89] | subcase of distribution, see [31] |
| distribution | [11, p. 101] | subcase of class *C* |
| class *B* | [1, Eq. (4) & (5)] | subcase of Demidenko |
| class *C* | [1, Eq. 6] | equivalent to *C*-property in [3, Sec. 4]; identical to class 2 in [2]; subcase of Demidenko |
| symmetric product | [25, Sect. 3], [11, Cor. 9] | subcase of symmetric increasing, see [25] |
| potential | [12, App.], [28, Sect. 2] | subcase of constant tour |
| convex planar | [3, Lemma 7] | subcase of convex-hull-and-line |
| edgeconvex | [15, Thm. 1] | subcase of Demidenko |
| convex 2-spherical | [3, p. 281] | subcase of edgeconvex |
| 3-line planar | [8] | subcase of *N*-line planar |

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